Skills Summary

Completing Short Algebraic and Segment Proofs

Use this reference while completing your worksheet or when studying.

A proof is a table of **statements** and **reasons**. The statements are the algebra you would have written anyway; the reasons say what gives you the right to write them. Line 1 is always the given, and the last line is always what you were asked to prove.

1. Justifying the steps of an equation

Properties of equality — one per move you make:

- add the same amount to both sides: *Addition Property*
- take the same amount off both sides: *Subtraction Property*
- divide both sides by the same nonzero number: *Division Property*
- clear parentheses: *Distributive Property*
- swap equal things for each other: *Substitution*

Both sides, every time: a move made on one side only is not a property.

Example: Given $4x - 9 = 23$, prove $x = 8$.
Step 1. Two moves separate x from the given equation, so the proof is two lines of algebra and the two properties that permit them.
Step 2. $4x = 32$ by the Addition Property (add 9 to both sides); then divide both sides by 4 by the Division Property.
$\boldsymbol{x = 8}$

Try it: Given $3(x + 2) = 21$, prove the value of x. Name a reason for every line.

check: $\boldsymbol{x = 5}$

2. Segment proofs with the addition postulate

Segment Addition Postulate: if B is between A and C, then $AB + BC = AC$. It is the only statement that lets you add segment lengths, so name it before the algebra starts. For four collinear points it applies twice.

Example: B is between A and C, $AB = x + 4$, $BC = 2x - 1$, $AC = 21$ cm. Find x.
Step 1. Betweenness is given, so the Segment Addition Postulate supplies the equation before any solving happens.
Step 2. $(x + 4) + (2x - 1) = 21$ gives $3x + 3 = 21$, then $3x = 18$ by the Subtraction Property and divide both sides by 3.
$\boldsymbol{x = 6}$ (check: $10 + 11 = 21$ cm)

Try it: Q is between P and R, $PQ = 5x$, $QR = 3x + 4$, $PR = 52$ cm. Find x.

check: $\boldsymbol{x = 6}$

3. Midpoints and congruent segments

Definition of midpoint: M is the midpoint of $\overline{AB}$ means $\overline{AM} \cong \overline{MB}$.

Definition of congruent segments: $\overline{AM} \cong \overline{MB}$ means $AM = MB$.

Two separate lines in a proof: congruence is about segments, equality is about their lengths, and only the second one can be solved.

Example: M is the midpoint of $\overline{AB}$ with $AM = 6x + 1$ and $MB = 4x + 11$ cm. Find x.
Step 1. Midpoint gives congruence and congruence gives equal lengths, so the equation to solve is $AM = MB$ — not $AM + MB$.
Step 2. $6x + 1 = 4x + 11$ gives $2x + 1 = 11$ by the Subtraction Property, then $2x = 10$.

$\boldsymbol{x = 5}$ (check: both halves are 31 cm)

Try it: N is the midpoint of $\overline{XY}$ with $XN = 7x - 2$ and $NY = 3x + 18$ cm. Find x.

check: $\boldsymbol{x = 5}$

(!) Watch out: "Combined like terms" is not a reason when the terms sit on opposite sides of the equals sign — that move is the Subtraction Property, and it has to happen on both sides.